ΕΛΛΗΝΙΚΗ ΔΗΜΟΚΡΑΤΙΑ
ΠΑΝΕΠΙΣΤΗΜΙΟ ΜΑΚΕΔΟΝΙΑΣ

Σχεδιαστικές Αποφάσεις

Ποιότητα Λογισμικού

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Outline

Design Decisions

Recap of Basic Metrics

Examples

Design options and decisions

A designer is faced with a series of design issues

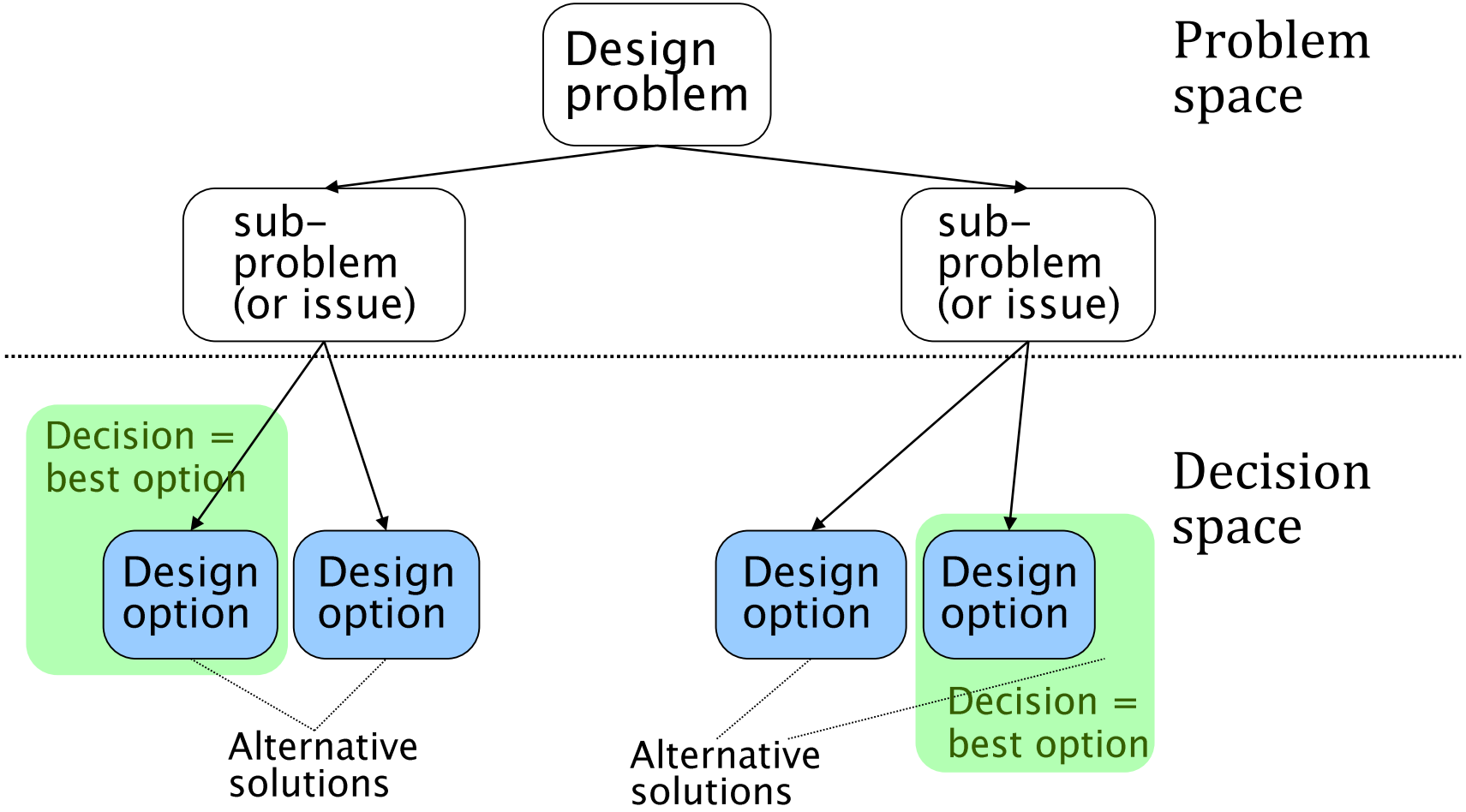
- These are sub-problems of the overall design problem

Each issue normally has several alternative solutions (or design options)

The designer makes a design decision to resolve each issue.

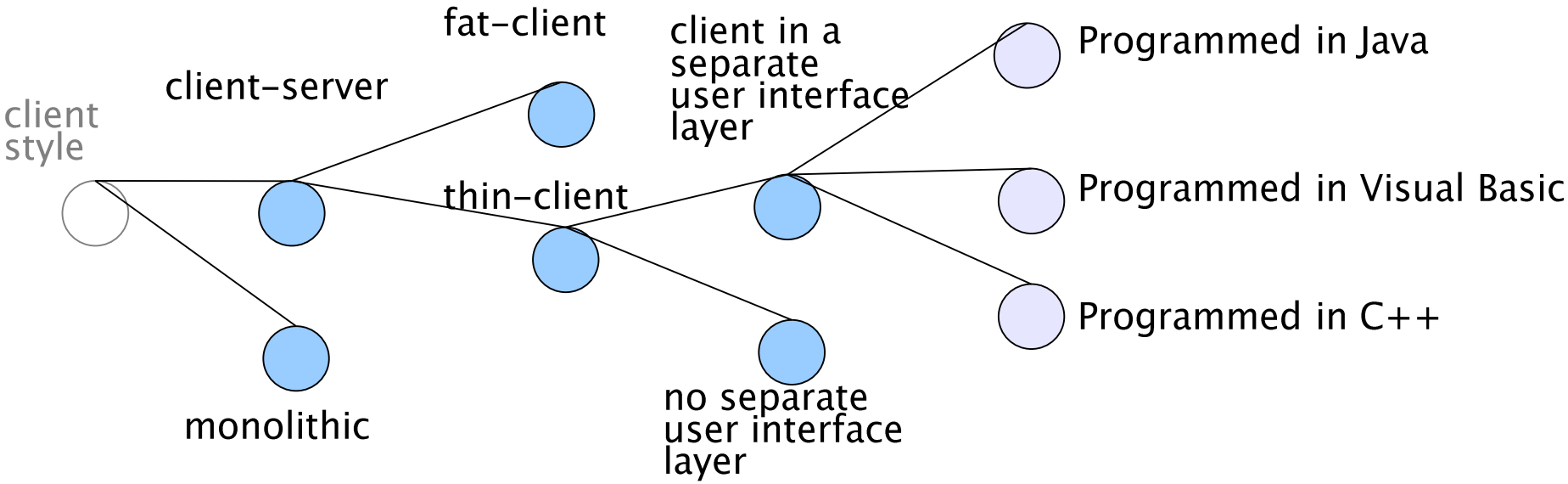
Choosing the best option from among the alternatives.

Taking decisions



Decision space

The space of possible designs that can be achieved by choosing different sets of alternatives.



Types of decisions

Implicit, undocumented

Unaware, tacit, of course knowledge

Explicit, undocumented

Vaporizes over time

Explicit, documented

Preferred, exceptional situation

Design Documentation

Prevents repeating (expensive) past steps

Explains why this is a good architecture

Emphasizes qualities and criticality for requirements/goals

Provides context and background

Elements of a design decision

Issues: design issues being addressed

Decision

- Status: e.g., pending, approved
- Assumptions: underlying assumptions

Alternatives

- Rationale; the why of the decision taken
- Implications: e.g. need for further decisions

Outline

Design Decisions

Recap of Basic Metrics

Examples

Object – Oriented Metrics

- Chidamber & Kemerer (1991)
  A suite of OO metrics
  Many variations

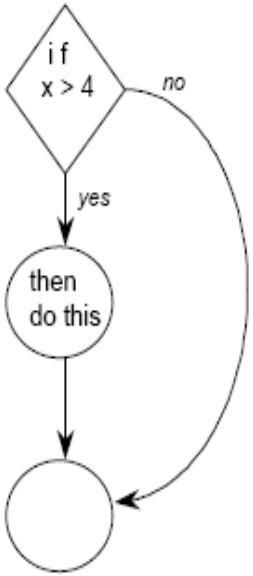
- Basic Categories:
  - Complexity
  - Coupling
  - Cohesion

Cyclomatic Complexity

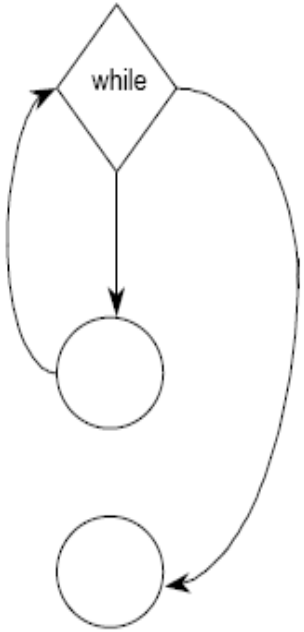
Cyclomatic Complexity

Number of Independent Test Paths => edges - nodes + 2

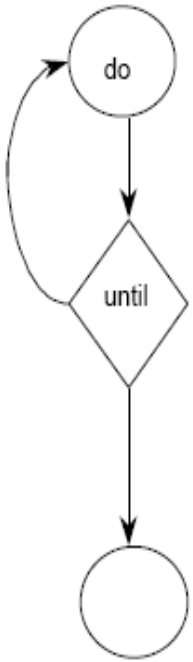
sequence:
1-2+2=1



if / then:
3-3+2=2

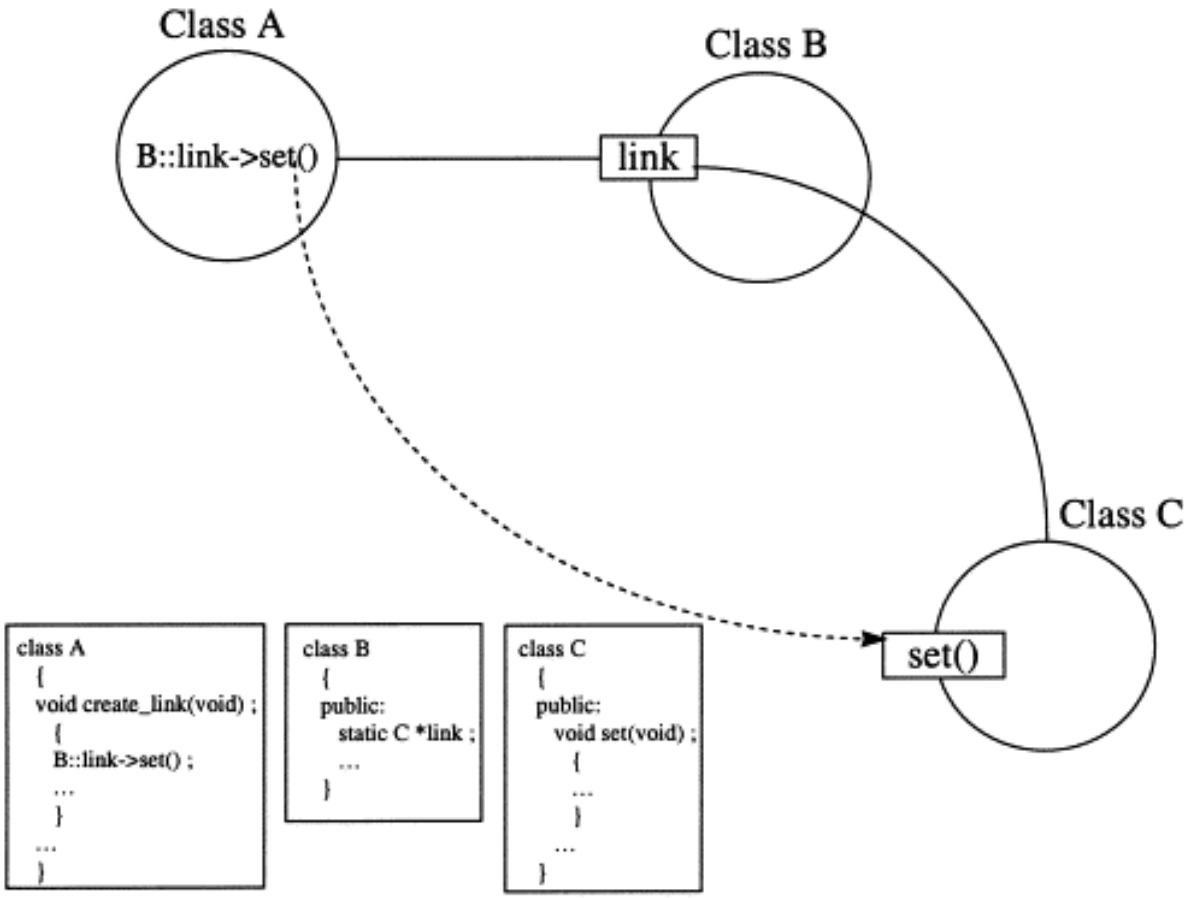


while loop:
3-3+2=2



until loop:
3-3+2=2

Coupling Between Objects



Lack of Cohesion of Methods

Let a class C with methods $M_1, M_2, \ldots, M_n$

Let $\{I_i\}$ the attributes used in M_i.

So there are several sets: $\{I_1\}, \{I_2\}, \ldots, \{I_n\}$

$$P = \{(I_i, I_j) \mid I_i \cap I_j = \varnothing$$

$$Q = \{(I_i, I_j) \mid I_i \cap I_j \neq \varnothing$$

$$LCOM = \begin{cases} |P| - |Q| & if\ |P| > |Q| \\ 0 & otherwise \end{cases}$$

Outline

Software Quality

Recap of Basic Metrics

Examples

Requirements-1

A software company that deals with software development and provides technical support

The company employs developers, analysts, managers and technical staff

All employees can be paid by hour or with a monthly salary

The company wishes to develop a system that will calculate the cost of all projects and the expenditures of the company for payroll.

Limitations-1

Cost estimation for software and technical support is calculated by a different algorithm that takes into account the amount of hours that each employee works on the project

Calculating the hourly payment of an employee is easy algorithm. On the other hand, the calculation of salary is more complex, since bonuses per project need to be calculated

Payment units are different across different types of employees

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Ερωτήσεις